

Wireless Networking

Question 1: List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

Answer:

Standard	Wi-Fi name	Frequency band	Max theoretical speed	Key difference
802.11n	Wi-Fi 4	2.4 GHz and 5 GHz (dual band)	Up to 600 Mbps	First standard to use MIMO and to support both 2.4 GHz and 5 GHz bands.
802.11ac	Wi-Fi 5	5 GHz only	Up to about 3.5 Gbps (multi-antenna models)	Works only on 5 GHz with wider channels (up to 160 MHz), giving much higher speed than 802.11n.
802.11ax	Wi-Fi 6 / 6E	2.4 GHz and 5 GHz (6E adds 6 GHz)	Up to 9.6 Gbps	Uses OFDMA, so it serves many devices at once with better speed in crowded places.

Question 2: Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

Answer:

The smartphone connects to the Wi-Fi router wirelessly using radio waves. The router is connected by a cable to the modem (or fibre ONT), which links the home to the Internet Service Provider (ISP) and the Internet.

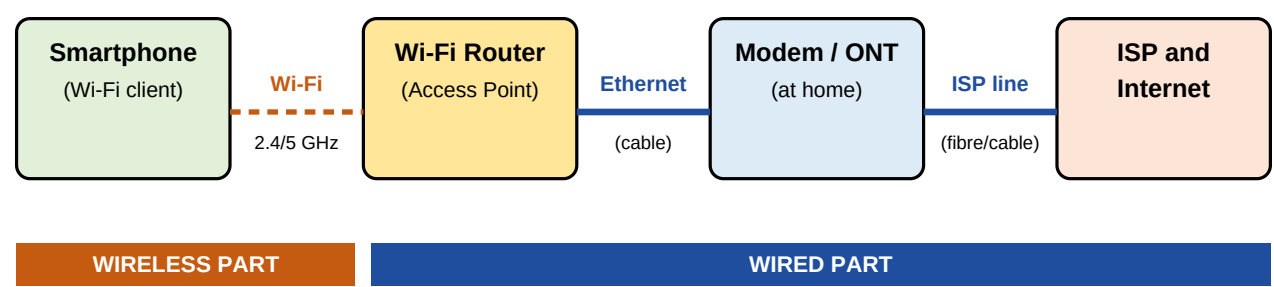


Figure 1: Smartphone connecting to the Internet through a home Wi-Fi router

Question 3: Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

Answer:

The Wi-Fi networks found during the scan are listed below:

S. No.	SSID	Signal strength	Security type
1	MyHome_5G	-42 dBm (Excellent)	WPA3-Personal

S. No.	SSID	Signal strength	Security type
2	College_Campus	-60 dBm (Good)	WPA2-Enterprise
3	Neighbour_WiFi	-72 dBm (Weak)	WPA2-Personal
4	Cafe_Free_WiFi	-55 dBm (Good)	Open (no security)

Choice: MyHome_5G. Signal strength is measured in dBm, and a value closer to 0 means a stronger signal, so -42 dBm is the strongest of the four and gives stable speed. It also uses WPA3-Personal, the newest and most secure option, which protects the password against offline guessing and encrypts each device's traffic separately. Cafe_Free_WiFi has a good signal but is open, so anyone nearby can capture the data sent over it. Neighbour_WiFi has both a weak signal and only WPA2, and I do not have permission to use it. College_Campus is secure, but it needs a college login. Therefore MyHome_5G gives the best combination of signal strength and security.

Question 4: Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

Answer:

What it is: The SSID (Service Set Identifier) is the name of a Wi-Fi network. Normally the router keeps broadcasting this name in small beacon frames so that nearby devices can show it in their list of available networks. SSID hiding (disabling SSID broadcast) stops the router from advertising the name, so the network does not appear in the list. To connect, a user must already know the exact SSID and type it manually, along with the password.

Advantage: The network is not visible to casual users, neighbours and passers-by, so it attracts fewer random connection attempts and looks less like an easy target.

Disadvantage: It does not give real security. The SSID is still sent inside other frames (such as probe requests and association requests) whenever a device connects, so it can easily be found with free tools like Wireshark. It is also inconvenient, because the name must be typed manually and devices keep searching for the hidden network, which can reveal its name in other places. Real protection comes from a strong password with WPA2 or WPA3 encryption.